

Prior Sensitivity Analysis framework for Bayesian Marketing Mix Models (MMM)

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Abstract

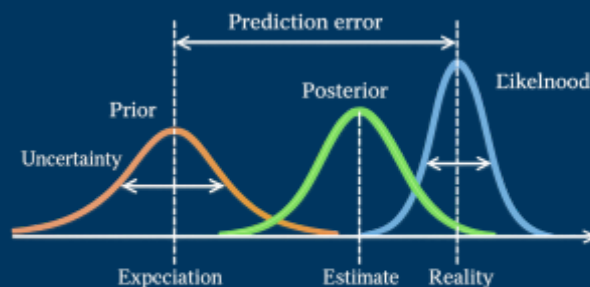
Our group develops a structured prior sensitivity analysis framework for Bayesian Marketing Mix Models (MMMs) to evaluate how modeling assumptions influence key marketing insights. Using modern Bayesian MMM approaches, including Google's Meridian framework, the project systematically affects prior distributions and measures resulting changes in outputs such as channel contribution estimates, return on investment (ROI), and forecasts.

By combining controlled prior variations with interpretable visualizations such as sensitivity summaries and tornado diagrams, the framework identifies which results are robust and which are highly dependent on prior assumptions. This work improves transparency and trust in Bayesian MMMs, particularly in privacy constrained environments where aggregated data is required, and supports more reliable decision-making for budget allocation and scenario analysis.

$$p(\theta | \text{data}) = \frac{p(\text{data} | \theta) \cdot p(\theta)}{p(\text{data})}$$

Posterior Likelihood Prior

Normalization



Background

The key ideas of the project includes:

Bayesian Statistics: Bayesian statistics is a framework for statistical inference in which probabilities represent **degrees of belief**. It combines prior knowledge with observed data using Bayes' theorem to produce updated beliefs (called posterior distributions).

- **Prior:** A prior contains **what is believed** about a model parameter before observing the data and is updated with data to form the posterior.
- **Prior Sensitivity:** Prior sensitivity refers to how much a Bayesian model's results **change** when different priors are used. If conclusions vary substantially across reasonable priors, the model is said to be prior-sensitive.

$$Y = \beta_0 + \beta_1 + \beta_2 X_2 + \dots \epsilon$$

Marketing Mix Models (MMM): Marketing Mix Modeling (MMM) is a statistical modeling framework used to measure how different marketing activities (e.g., TV, search ads, social media, promotions) **impact** business outcomes.

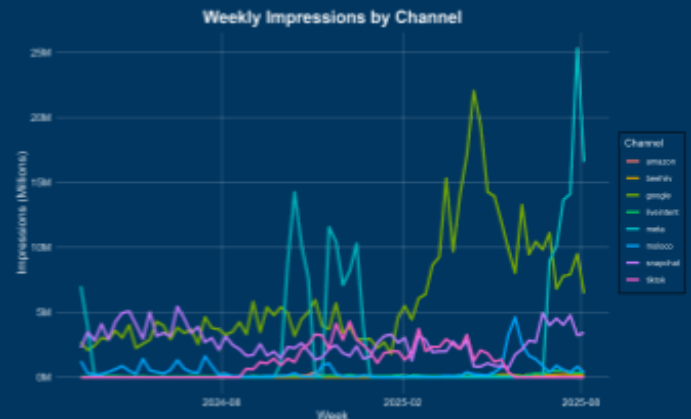
- **Google Meridian:** Google Meridian is Google's **open-source Bayesian MMM** framework. It provides privacy-safe measurement of marketing effectiveness by combining Bayesian modeling, prior sensitivity analysis, and causal reasoning.
- **Causal Reasoning:** Causal reasoning is the process of identifying and estimating the **cause-and-effect relationships** between typical marketing actions (e.g., advertising spend, promotions) and business outcomes (e.g., sales or conversions).

Data

The dataset consists of weekly aggregated marketing data, with one observation per week, linking marketing activity to business outcomes over time. It includes platform-level inputs such as ad spend and impressions across multiple digital channels, along with revenue and conversion based outcomes including subscriptions, purchases, and ROI metrics. The data represents complete historical records collected from advertising reporting systems and is observational rather than experimental, meaning causal interpretation relies on modeling assumptions rather than randomization.

Objectives

1. Build a reproducible prior sensitivity framework for Bayesian MMMs, referencing Google Meridian.
2. Evaluate how prior assumptions affect channel contribution, ROI, and forecasts.
3. Analyze the MMM outcomes and Identify which conclusions are robust versus prior-sensitive.
4. Apply the framework using modern MMM tools.
5. Create clear visual and summary outputs.
6. Support transparent and reliable decision-making in privacy-constrained settings.



Current Progress

We currently review the structure of Bayesian Marketing Mix Models and identify key prior distributions that influence model behavior, such as priors on channel effects and saturation parameters. We are familiarizing ourselves with modern Bayesian MMM tools, including Google Meridian, and discussing how prior assumptions are specified and interpreted in practice.

Next Step

- Implement a reproducible prior sensitivity analysis pipeline.
- Run controlled experiments across a range of reasonable prior choices.
- Quantify how key MMM outputs change in response to prior variations.
- Summarize results using interpretable tables and visualizations.
- Clearly communicate robustness versus sensitivity of model conclusions.